

Conditional Per-Step Budget Certificates

A Receipt-Grade Interface for Adaptive Horizon Claims (Series Synthesis)

Unified archive note v0.05 (2026-02-28; archive v123)

Abstract

The archive publishes privacy budgets as compact receipt line items with digest-bound replay hooks. For repeated use (retries, fallbacks, and interactive protocols), additive horizon claims do *not* require independence, but they *do* require a conditional per-step guarantee. This note defines a small, auditable object shape—a *conditional per-step budget certificate*—that is sufficient to justify horizon- n receipts under adaptive interaction. Other papers should cite this note instead of re-explaining what “replayable per-step conditional bounds” means.

Scope. This note is an interface: it defines what must be committed to (and replayable) for Regime II horizon accounting. It is not a new privacy theorem; the sequential-composition proof is owned by Anonymity A.[1]

Imports. Receipt line-item fields (including `replay_hook`) are owned by Synthesis 12.[2] Horizon-regime rules (I/II/III) are owned by Synthesis 19.[3] Plan/state drift semantics for digest-bound hooks are owned by Synthesis 18.[4] The mechanical replay checklist is owned by Synthesis 20.[5]

1 Why this certificate exists

Anonymity A gives a sequential-composition theorem: if each step satisfies a conditional one-vs-mixture bound given the public history, then the total odds-inflation (equivalently, pointwise MaxL in bits) is bounded by the sum of per-step budgets.[1] Receipts want to publish horizon claims (“ $Q\times$ ”) without assuming independence; this is Regime II in Synthesis 19.[3] To keep papers short and referee-grade, we isolate the *auditable commitment* that makes the theorem applicable.

2 Conditional per-step budget certificates

We write I for the hidden identity (or secret) and O_t for the public output at step t . Let H_{t-1} denote the public history (all public outputs so far, plus the declared schedule/plan information).

Definition 1 (Conditioning interface). *A conditional certificate must specify a public conditioning interface: a deterministic projection*

$$S_{t-1} \stackrel{\text{def}}{=} g_t(H_{t-1})$$

together with a digest identifier `cond_iface_id` that binds the code/specification of g_t . Intuitively, S_{t-1} is the public summary of history that the per-step guarantee conditions on.

Definition 2 (Conditional per-step MaxL certificate). *Fix a step t . A conditional per-step MaxL certificate for t consists of:*

1. *a bound value $b_t \geq 0$ (in bits, i.e. $\log_2$ units),*
2. *a conditioning interface `cond_iface_id` (Definition 1),*
3. *a digest-bound evidence hook `evidence_id` that enables an auditor to check the inequality below for the realized conditioning value S_{t-1} .*

The certificate asserts that, almost surely over the public history, the realized output satisfies

$$\max_i \log_2 \frac{\Pr[O_t = o \mid I = i, S_{t-1}]}{\Pr[O_t = o \mid S_{t-1}]} \leq b_t \quad \text{for the realized } o = O_t. \quad (1)$$

The evidence hook may be implemented by (i) a static dual/cut certificate (when the kernel is history-invariant), (ii) a finite table over conditioning states, or (iii) a replayable program committed by `plan_spec_id` that deterministically computes the checked inequality from public logs and requested artifacts.

Remark 1 (A minimal record shape for tool output). *When tooling emits per-step certificates (for use inside a `plan_spec`), a compact record format is:*

$$\{ \text{step_selector}, \text{cond_iface_id}, \text{bound_bits}, \text{evidence_id} \}.$$

Here `step_selector` names which step(s) the bound applies to (a single t , a range, or a predicate on public schedule time), and `evidence_id` points to either (i) a static dual/cut certificate, (ii) a finite conditioning table, or (iii) a log-audit object. The equalization-audit lemma in Anonymity B is an example of a log-audit evidence object that can populate `evidence_id` for a discrete projection.[9]

Remark 2 (Where this lives in a receipt). *Receipts do not need new fields. A line item that claims a horizon- Q total under Regime II must include a `replay_hook` (`plan_id`, `plan_spec_id`) whose plan specification commits to: (i) the conditioning interface `cond_iface_id`, (ii) the per-step bounds b_t (or a deterministic rule that yields them), and (iii) the evidence hooks `evidence_id` needed for replay. This is an instance of the general replay discipline in Synthesis 20.[5]*

Proposition 1 (Receipt-facing sufficiency). *If a receipt’s replay hook provides conditional per-step MaxL certificates (Definition 2) for steps $t = 1, \dots, T$, then the horizon- T transcript satisfies a total MaxL/odds-inflation budget of at most $\sum_{t=1}^T b_t$ bits.*

Remark 3 (Stateful deployments). *For anonymous DHTs, routing tables, caches, congestion signals, and scores can make S_{t-1} large. Regime III in Synthesis 19 treats this by enlarging the witness family via a digest-bound `state_decl_id`. [3, 4] A Regime II certificate remains usable in the stateful setting as long as the conditioning interface g_t is defined over the declared state surfaces (and the replay plan obtains the relevant state readouts from the transparency bundle).*

3 Research warranted (kept out of other papers)

Two recurring barriers to small certificates are (a) huge conditioning spaces and (b) optional stopping / drift monitoring. For (a), a promising direction is to certify contraction of the conditioning channel (e.g. via a spectral gap or a mixing-rate certificate) so one can bound worst-case conditional leakage by a small state proxy.[6] For (b), Operational B’s anytime-valid evidence suggests “drift certificates” that remain safe under optional stopping.[8] We keep those research threads out of mechanism papers; this note is the single citable landing place.

3.1 Spectral contraction as a compact evidence object (pointer)

One concrete way to keep Regime II certificates small is to certify *contraction* of a delegation kernel rather than enumerate a huge conditioning space. For random-walk delegation, the archive’s published spectral anonymity note provides sharp distinguishing bounds and conservative certificates (e.g. via a second-singular-value bound).[6]

This note does *not* re-derive spectral translations. The canonical “spectral witness $\Rightarrow$ per-step MaxL cap (bits)” recipe is Anonymity D.[7] Mechanism papers that want to export a Regime II per-step certificate from a delegation walk should cite Anonymity D for the numeric translation and cite this note for the receipt-side certificate shape.

References

- [1] Anonymous. Odds-Inflation Anonymity: Posterior Odds Budgets, Composition, and Operational Limits. Anonymity series Paper A, The-One-True-Archive (2026).
- [2] Anonymous. Receipt Line Items for Anonymity Claims: A Minimal Tuple Schema for Published Budgets. Synthesis 12, The-One-True-Archive (2026).
- [3] Anonymous. Receipt Accounting and Horizon Composition for Anonymous-DHT Budgets: A Rulebook for Repeats, Retries, and State. Synthesis 19, The-One-True-Archive (2026).
- [4] Anonymous. Plan and State Equivalence: Digest Binding, Drift, and Replay Semantics. Synthesis 18, The-One-True-Archive (2026).
- [5] Anonymous. Auditor Replay for Anonymous-DHT Receipt Line Items: A Mechanical Checklist for Digest-Bound Claims. Synthesis 20, The-One-True-Archive (2026).
- [6] Anonymous. Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlays. Published note (2026-01-23), The-One-True-Archive (2026).
- [7] Anonymous. Spectral Delegation for Anonymous DHT Lookups: From mixing certificates to receipt-grade per-step budgets. Anonymity series Paper D, The-One-True-Archive (2026).
- [8] Anonymous. Auditable Trace Privacy with Abort: Disclosure receipts for side-channel resistant transport. Operational series Paper B, The-One-True-Archive (2026).
- [9] Anonymous. Anonymous DHT Profiling and the Equalization Contract. Anonymity series Paper B, The-One-True-Archive (2026).